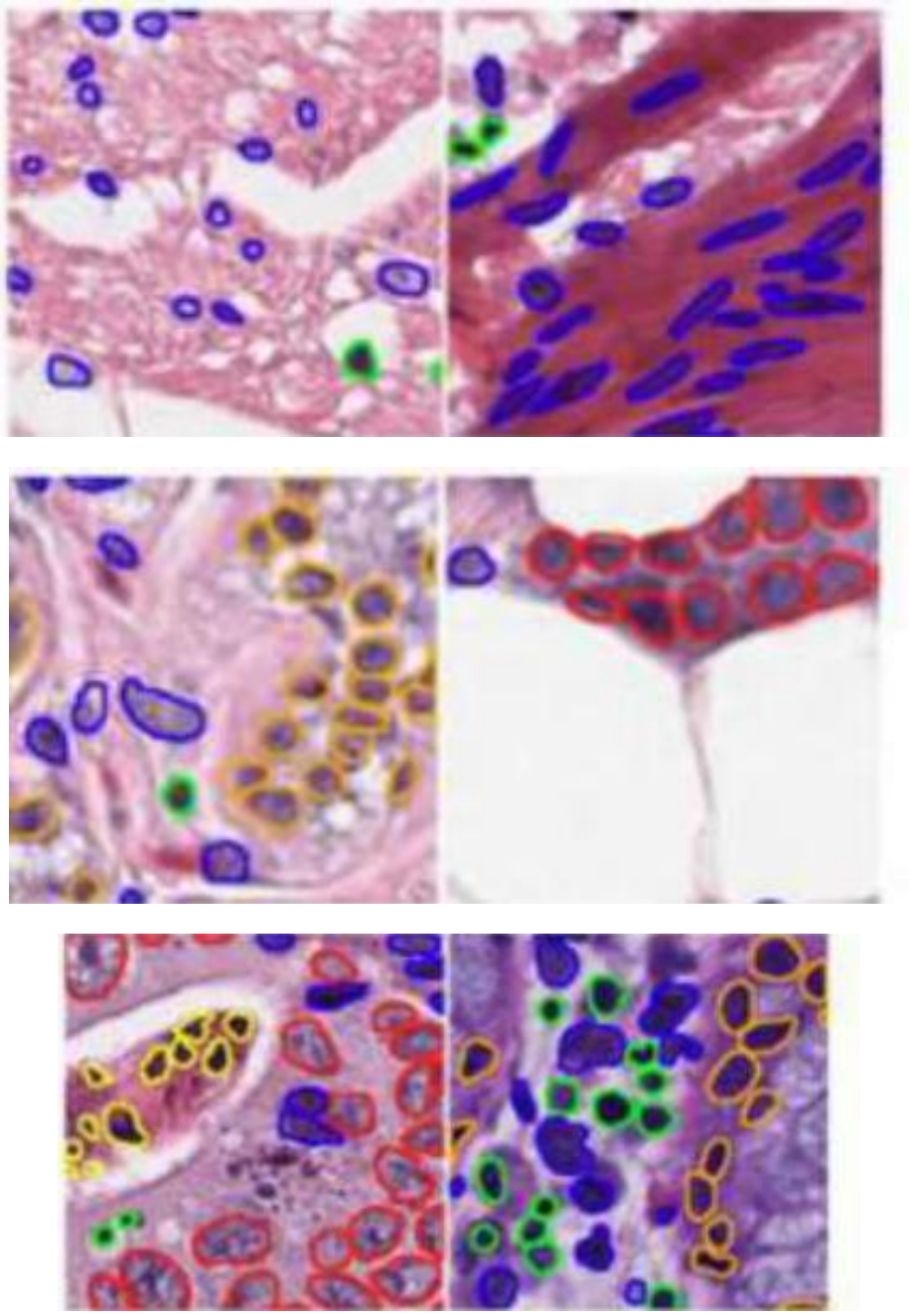
BIOMEDICAL DATA DESIGN I

Nuclei instance segmentation across diverse tissues

HoVer-Net, CellViT-UNI, and PM-Net

Weekly report • October 15, 2026

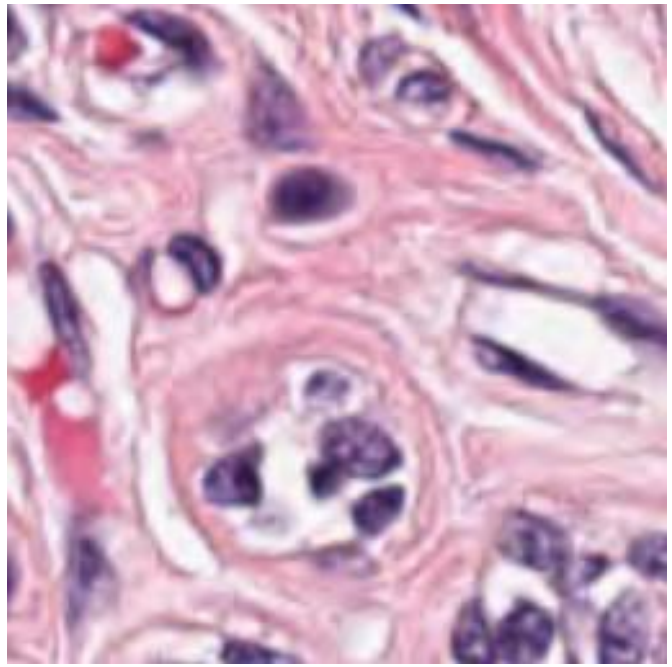
Xinhao Jin



THE TASK

A nucleus needs its own mask and class label

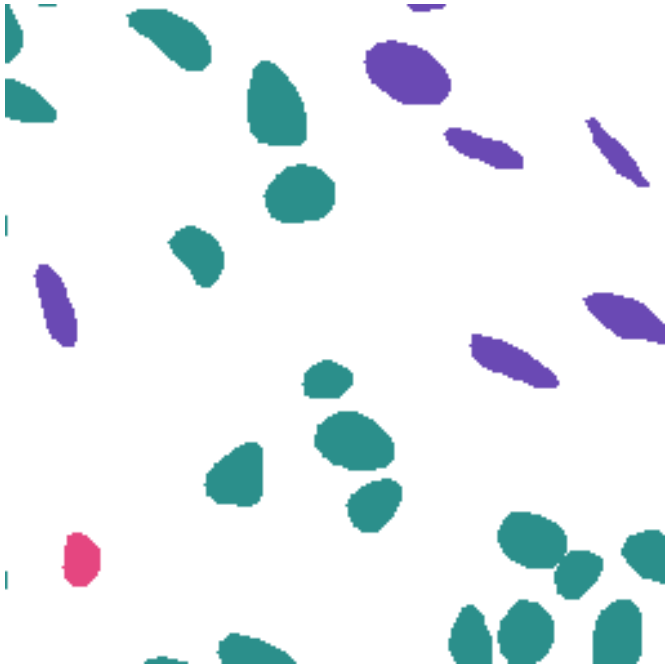
H&E image



Individual instances



Five nucleus classes

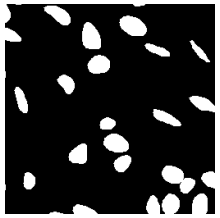


Touching nuclei must remain separate objects.

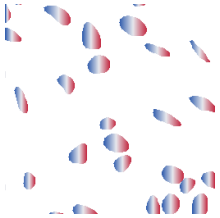
One shared encoder feeds three decoder branches

CNN encoder ResNet-50

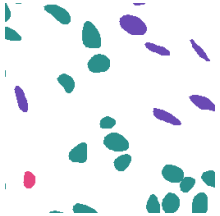
NP: nucleus pixels



HV: relative position



TP: nucleus type



Separate jobs,
shared image features

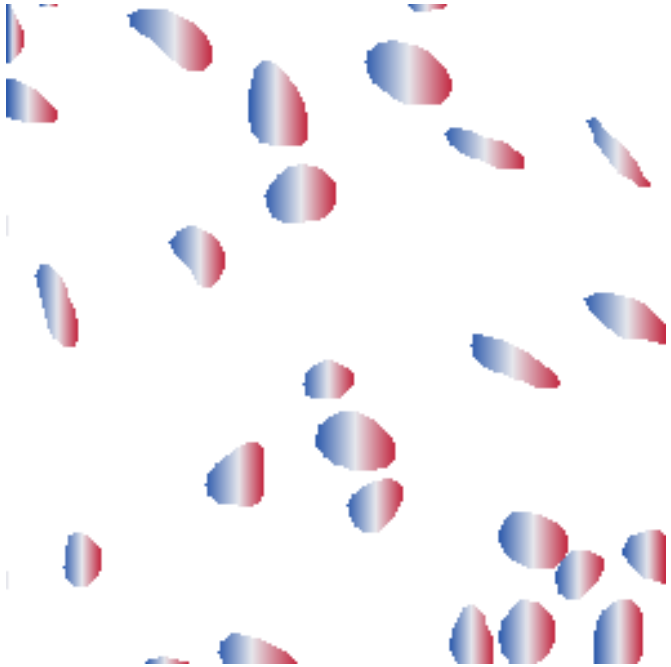
NP and TP are pixel classification. HV predicts two continuous values.

Each nucleus has its own horizontal and vertical coordinates

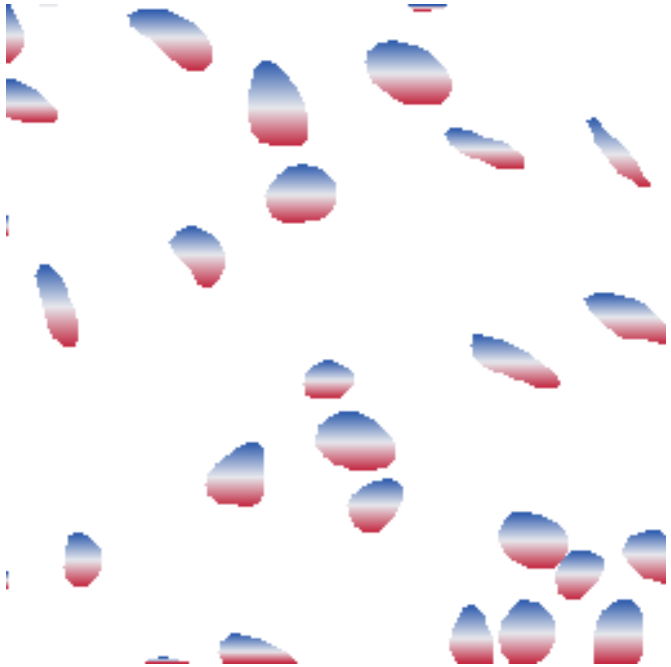
Instance IDs



Horizontal offset

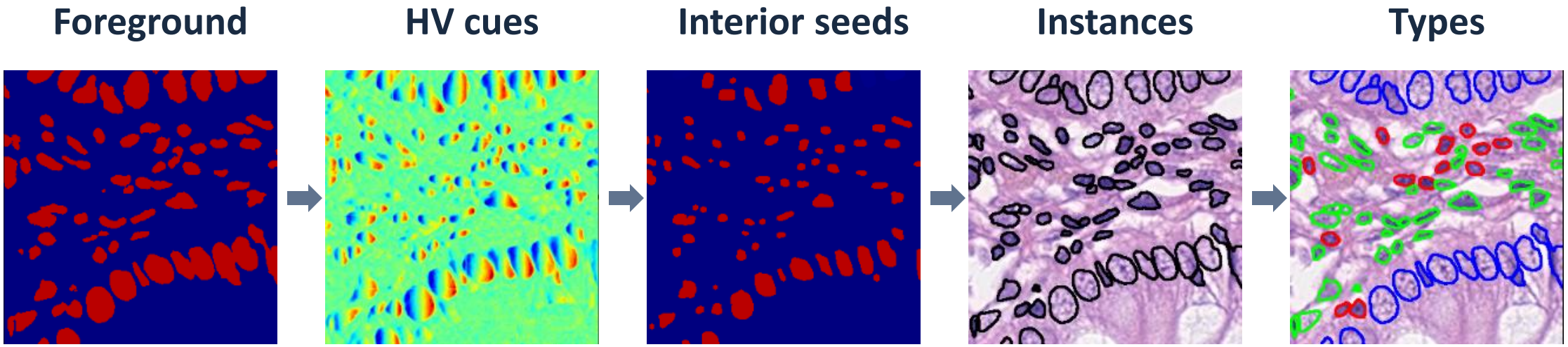


Vertical offset



Blue = negative offset. Red = positive offset. White background is outside nuclei.

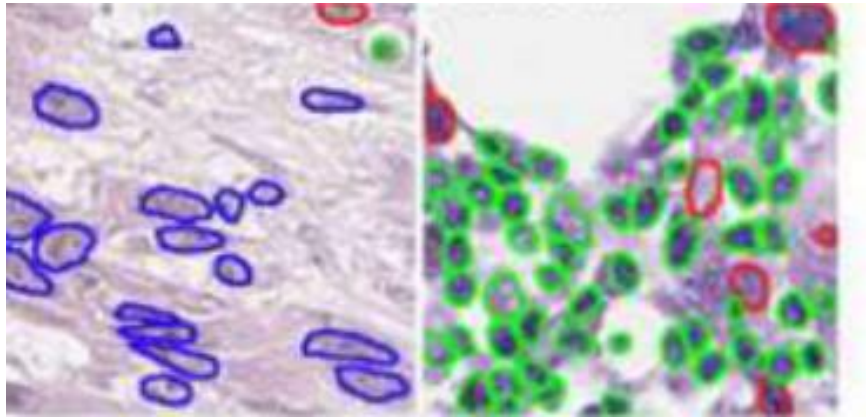
Watershed turns pixel maps into separate objects

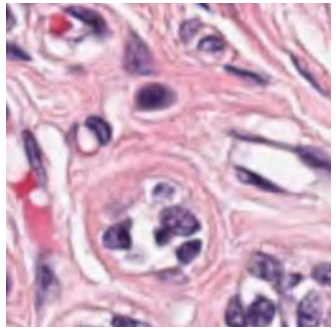


Original prediction and decoding panels from Graham et al. (2019), Fig. 1.

Missing seeds can merge nuclei. Extra seeds can split one nucleus.

UNI learns pathology features before our task training





More than 100 million image patches

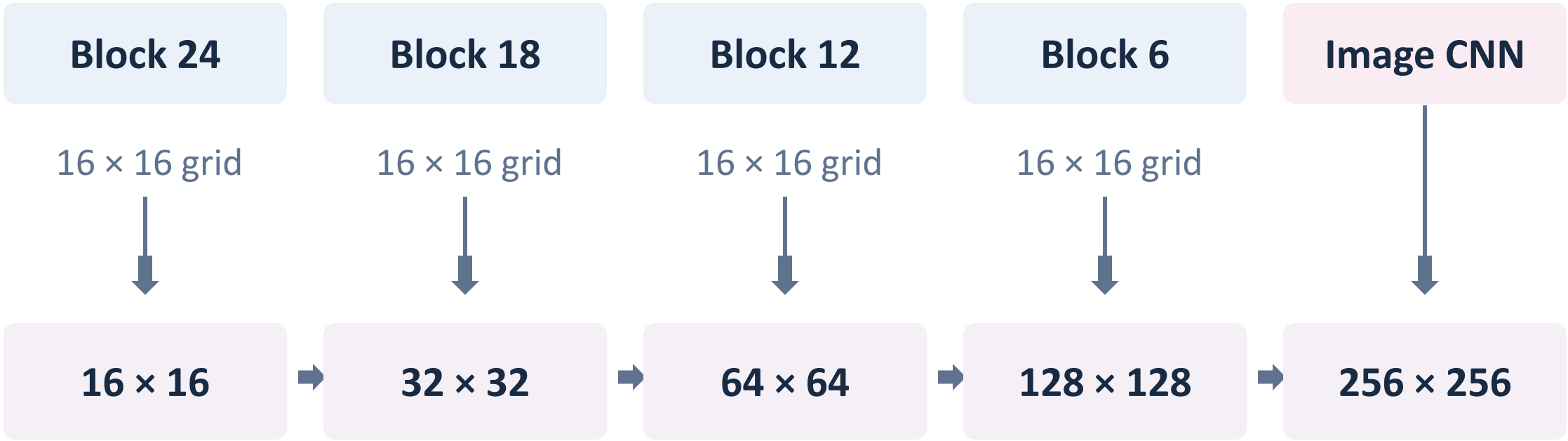
Over 100,000 H&E whole-slide images

Self-supervised pretraining
across 20 major tissue types

UNI supplies image features. The nucleus decoder still has to learn our task.

CELLVIT DECODER

Features from four depths recover fine spatial detail



Upsampling restores resolution; skip features supply detail at each stage.

BASELINE EVIDENCE

CellViT-UNI improves mPQ, but Dead misses remain

Own training, no TTA	mPQ	bPQ
HoVer-Net	0.4564	0.6613
CellViT-UNI	0.4995	0.6653

38.5%

of Dead nuclei were missed in the original baseline diagnosis

Three-split means above. The miss-rate diagnosis is a separate historical analysis.

What the completed studies tell us

Route	What we tested	Result and interpretation
Typing and data	CONCH, T0–T3, copy-paste, diffusion, CF	No stable gain. Retyping cannot recover missed nuclei.
Models and scale	HoVer-NeXt, LKCell, KongNet, P0–P3	mPQ 0.4579 / 0.4923 / 0.3460. More scale adds errors.
Mechanism checks	E2a, DSB, E1, M, E3	No sufficient HV or conflict explanation. Label uncertainty remains.
Cost and supervision	EF-P2 / E0, E4, four-arm study	Small scale gain is costly. Selective HV fails its criteria.

No completed route has shown a clear overall improvement.

Filtered scale additions keep a small Dead gain at high cost

Nine matched split / seed pairs	mPQ	bPQ	Dead PQ
Matched baseline	0.49760	0.66530	0.17255
P2 additions	0.49769	0.66443	0.17997
EF-P2 filtered additions	0.49828	0.66529	0.17954

+0.0070 Dead PQ

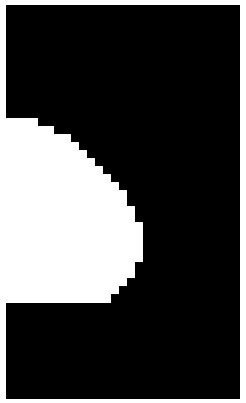
≈6.1× inference cost

Image-bootstrap intervals span zero. Historical test folds were reused across rounds.

Does changing HV targets repair crop-related misses?



Visible nucleus



Cropped mask

E4: cropped Dead nuclei were missed more often.

Arm	Crop prob.	HV minimum area
A0	0.1	30 pixels
A1	0.5	30 pixels
A2	0.5	1 for new small crop fragments
A3	0.5	1 for all objects

A2 – A1 isolates selective HV supervision from stronger cropping.

SUPERVISION RESULTS

The original-image quality criteria fail

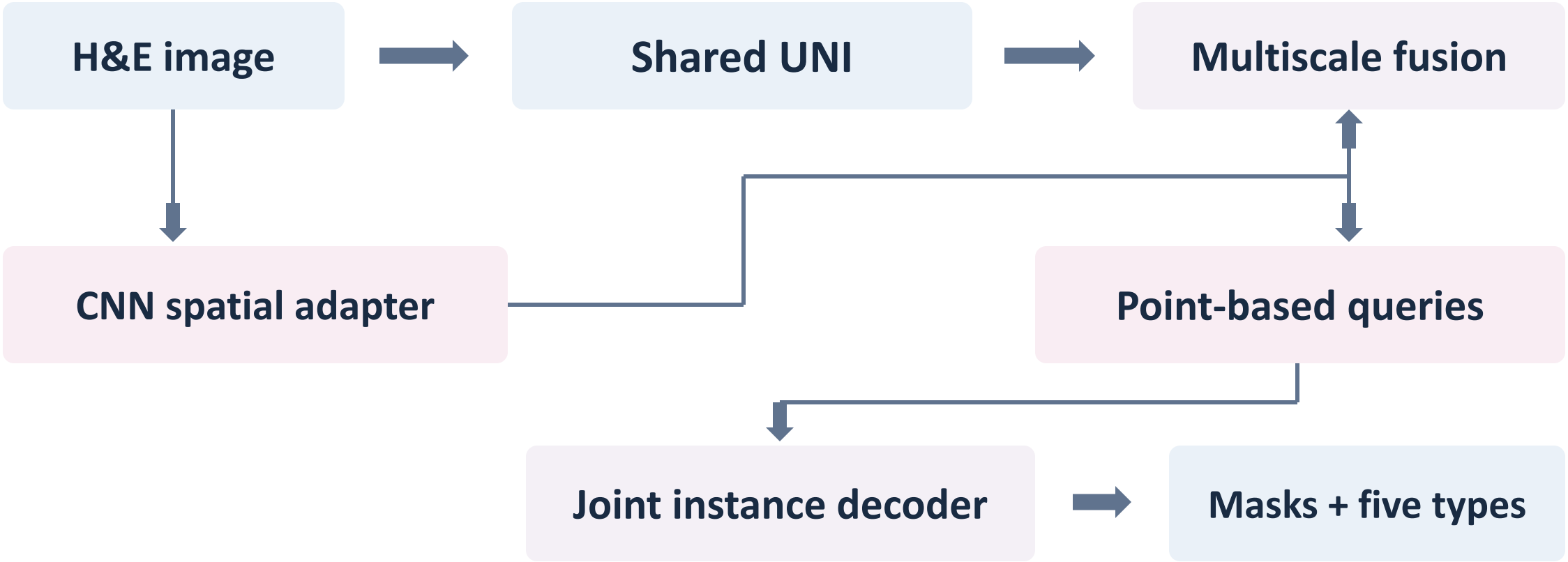
Mean change, two seeds	mPQ	bPQ	Dead PQ	Strict Dead
A1 − A0	−0.0053	−0.0061	+0.0376	+0.0088
A2 − A0	−0.0063	−0.0054	+0.0246	−0.0009
A2 − A1	−0.0010	+0.0007	−0.0131	−0.0098

Selective HV adds no Dead benefit over crop-only training.

Validation only. A2 exceeds the 0.002 mPQ / bPQ loss tolerance in both seeds.

PM-NET ARCHITECTURE

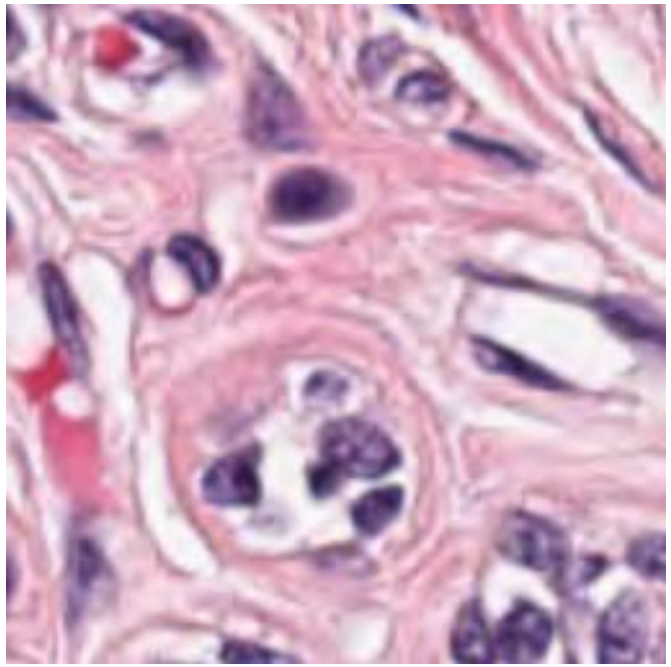
One image encoding supports all nucleus queries



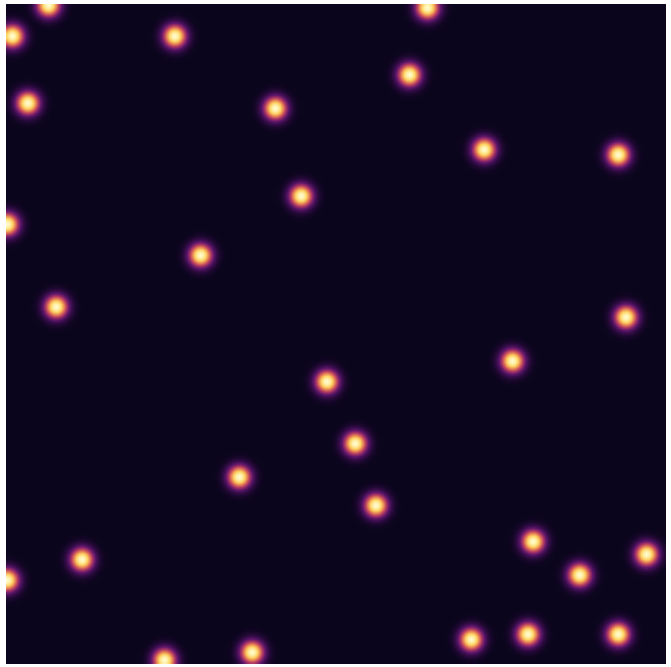
Proposed model. No HV map or watershed in the instance decoder.

A point target must stay inside the visible nucleus

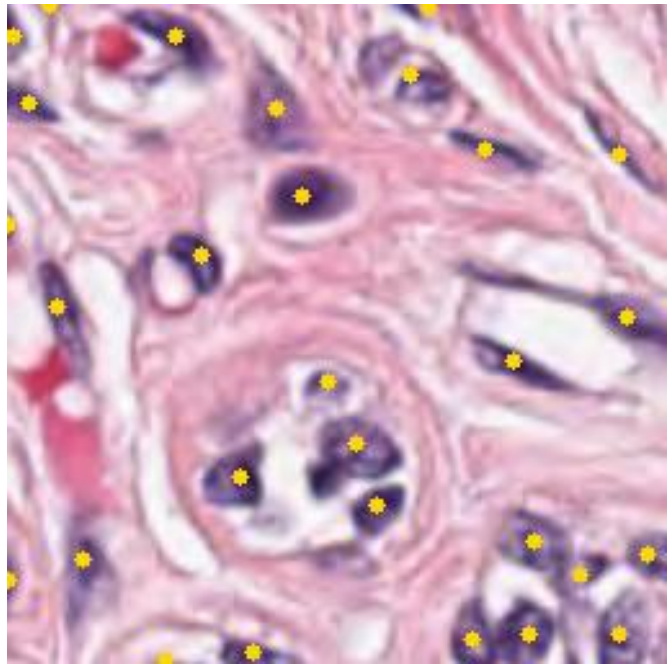
Image



Illustrative point target



Inside-mask points



Direct heatmap supervision trains the point head despite discrete point selection.

Point initialization and fusion must earn their added cost

Comparison	Question
Basic queries vs point-initialized queries	Does point initialization help?
Plain vs gated multiscale fusion	Does the fusion rule help?
Parameter-matched ordinary adapter	Is the gain only extra capacity?

276 objects in one training patch

256 queries are insufficient. 512 covers original labels, but cost is untested.

Also handle 23 rounded centers outside masks and 116 empty-label training images.

EVALUATION PLAN

Detection gains must preserve quality and inference cost

Endpoint	Proposed development requirement
Interior Dead recall	Mean gain ≥ 0.03; paired lower 95% bound > 0
Dead PQ	Mean gain ≥ 0.005; both seeds improve
mPQ / bPQ	Per-seed loss no more than 0.002
Strict scores and instance errors	No strict decline; mean merge/split rates do not rise
End-to-end latency	At most 2× the single-scale baseline

Matched UNI initialization, schedule, seeds, final checkpoints, and evaluator.

NEXT STEPS

A staged plan leaves a useful result even if PM-Net fails

1	**Minimum**	Baselines, error analysis, and completed supervision results
2	**Target**	A tested PM-Net prototype with two-seed ablations
3	**Stretch**	Three-split and external confirmation if dev criteria pass

The next study tests the instance representation, with clear quality and cost limits.

Detection, mask quality, and type all affect the score

mPQ	**Type-aware quality**	Class-specific PQ, averaged across five classes
bPQ	**Instance quality**	All nuclei treated as one foreground class
Dead PQ	**Rare-class quality**	Focus on dead nuclei
Strict scores	**Absent-class penalties**	Retain false positives when a class is absent

An official Dead gain can coexist with worse strict Dead or bPQ.

HV area thresholds and class conflict are incomplete explanations

E2a: scale × HV target support, four arms and two seeds

Lowering the x1 cutoff reproduces only 9.6% of the x2 Dead gain.

DSB: test the need for a separate Dead expert

Mean gradient cosine +0.749; none of 64 valid batches is negative.

Decision

Keep HV cutoff 30. Stop the specialist route before expert training.

The measured class-conflict premise failed under the tested conditions.

Detector agreement is useful evidence, but not ground truth

E1-C / E1-T: agreement across detectors

66–73% candidate support; 72.5% vote for the base model's disputed type.

M: merge inflammatory and connective labels

Official mPQ changes by +0.0519 in a different label space.

E3: learn candidate utility

AUC 0.721 / 0.737, but strict Dead criteria fail. Identity fallback.

Expert review remains unavailable. Five-class method scores are unchanged by M.

The crop-mechanism criterion also fails

–0.0086

Mean A1 – A2 miss-rate improvement

95% interval: [–0.0409, +0.0177]

116 eligible Dead objects in 38 images; inference covered 2,523 fixed windows.

Stop this dev recipe. No extra seeds or three-split confirmation.